MILITARY SPECIFICATION

FUZE, POINT DETONATING, M505A3; PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for use in Fuze, Point Detonating (PD), M505A3. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)

MILITARY

MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Ammunition Lot Numbering

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Philadelphia, Pennsylvania 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 (Cont)

MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMAMENT COMMAND

D7258864	- Metal Parts and Metal Parts Assembly
LI 7258864	- Index of Inspection Equipment Lists for: Fuze PD, M505A3, Parts for

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. -The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117-73 -- Method of Test for Salt Spray (Fog) Testing
ASTM E18-67 -- Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General.- The metal parts and the metal parts assembly shall be in accordance with the requirements of Drawing (Dwg.) D7258864, referenced specifications and other requirements specified herein.

3.2 Manufacturing process. -All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.4)

3.3 First article sample.- Unless otherwise directed by the contracting officer, a first article sample shall be required.

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3.4 Hardness, rotor detent spring.-The hardness of the rotor detent spring shall be as specified by Dwg. C7258869, a detail of Dwg. D7258864.

3.5 Protective coating.

3.5.1 Plating. -The plating on the booster holder, body and nose cap shall be in accordance with the requirements of the applicable drawings and QQ-P-416.

3.5.1.1 Corrosion resistance. -The exterior plated surface of the booster holder and the metal parts assembly shall show no white corrosion products of cadmium or corrosion of basis metal after exposure to a 5 percent solution of atomized salt spray for the time specified by the applicable drawing.

3.6 Workmanship. -The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.6.1 Processing defects.-The metal parts and the metal parts assembly shall be free of dents, cuts, deep scratches, unbroken and ragged edges, slivers, burrs, damaged plated surfaces and other forms of substandard workmanship.

3.6.2 Cleanliness. -The metal parts and the metal parts assembly shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. -Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of inspections. -The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection.

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4.3.1 First article sample.- The sample shall be delivered for inspection in accordance with first article approval provisions of the contract and MIL-A-2550, and shall consist of:

50 - Bodies
50 - Nose caps
50 - Firing pins
50 - Rotor detent springs
50 - Rotors
50 - Booster holders
50 - Metal parts assemblies

4.3.1.1 Inspection.- After inspection by the contractor and provisional acceptance by the Government activity, the sample will be inspected in accordance with the provisions of MIL-A-2550. Failure of the original first article sample to be in accordance with the applicable drawings and the requirements of Para. 3, as specified herein, will result in sample disapproval. (see 6.3)

4.4 Quality conformance inspection.

4.4.1 Submission of Product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot.- A lot shall consist of metal parts produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.4.1.2 Lot identification.- Each lot of fuze metal parts shall be identified in accordance with MIL-STD-1168 and packaged and packed in accordance with MIL-STD-1169, supplemented as directed by the procuring activity.

4.4.2 Examination.- Examination for major and minor defects shall be performed by the contractor as specified herein. Each part and each assembly found to be defective shall be rejected.

4.4.2.1 Sampling and acceptance criteria.- The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.2.c) In addition, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.5)

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4.4.2.2 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.25%
Minor	0.40%

b. Class basis:

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis:

Major	0.40%

4.4.3 Classification of defects 1/.-The The classification of defects shall be as follows: (see 6.6)

4.4.3.1 Body (See Dwg. D7258866, a detail of Dwg. D7258864).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Pitch diameter (P.D.), external thread	Gage
102. P.D., internal thread	Gage
103. Position, P.D., internal thread with datum surfaces	Gage
104. Position, spherical surface, with datum surfaces	Gage
105. Depth to top of precision ball	Gage
106. Depth to rotor detent spring groove	Gage
107. Diameter, internal cavity rear of rotor detent spring groove	Gage
108. Diameter, firing pin guide hole	Gage
109. Position, firing pin guide hole with datum surfaces	Gage
110. Break, perimeter of firing pin hole, max.	Gage
111. Length, base to shoulder	Gage
112. Depth to end of full internal thread, min,	Gage
113. Foreign matter	Visual 2/
114. Diameter, rotor detent spring groove	Gage
Minor:	
201. Major diameter, external thread	Gage
202. Minor diameter, internal thread	Gage
203. Profile, shoulder to nose, except tapered surface forward of crimping groove	Gage
204. Runout, body taper with datum surfaces	Gage
205. Width, rotor detent spring groove	Gage

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4.4.3.1 (Cont)

Categories and Defects	Method of Inspection
Minor: (Cont)	
206. Diameter, thread relief at shoulder and base	Gage
207. Width, thread relief at base; max.	Gage
208. Width, thread relief at shoulder, max.	Gage
209. Wall thickness variation, rotor detent spring groove, max.	Gage
210. Internal thread relief missing	Visual
211. Finish, interior	Visual/Feel
212. Sharp corner or edge	Visual/Feel

4.4.3.2 Nose cap (See Dwg. C7258867, a detail of Dwg. D7258864).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Thickness, datum area	Gage
102. Depth, to spherical surface	Gage
103. Length, crimping lip, min.	Gage
104. Foreign matter	Visual
Minor:	
201. Length, crimping lip, max.	Gage
202. Angle, crimping lip with conical surface	Gage
203. Inside diameter, crimping lip	Gage
204. Profile, closed end	Gage
205. Thickness, except datum area	Gage
206. Burr or sliver, inside edge of crimping lip	Visual/Feel

4.4.3.3 Firing pin (See Dwg. C7258868. a detail of Dwg. D7258864).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Total length	Gage
102. Thickness, flange	Gage
103. Diameter, body	Gage
104. Profile, point	Gage
105. Damage	Visual
106. Foreign matter	Visual

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4.4.3.3 (Cont)

Categories and Defects	Method of Inspection
Minor:	
201. Profile, head	Gage
202. Diameter, flange	Gage
203. Length, flange to shoulder	Gage
204. Radius, at intersection of flange and body, max.	Gage
205. Finish	Visual/Feel
206. Burr or sharp edge	Visual/Feel

4.4.3.4 Rotor detent spring (See Dwg. C7258869, a detail of Dwg. D7258864).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Inside diameter (I.D.), max.	Gage
102. Width, at center line of throat radius	Gage
103. Deformity (excluding out of flatness)	Visual
104. Foreign matter, including corrosion	Visual
105. Burr or sharp edge	Visual/Feel
Minor:	
201. Outside diameter	Gage
202. Position, I.D. with datum surface	Gage
203. Thickness, min., and including out of flatness, max.	Gage
204. Width, across opening	Gage
205. Radius, opposite open end	Visual
206. Finish	Visual/Feel
207. Radii, open end	Visual 3/

4.4.3.5 Rotor (see Dwg. D7258870, a detail of Dwg. D7258864).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Spherical diameter	Gage
102. Diameter, detonator cavity	Gage
103. Depth, detonator cavity	Gage
104. Diameter, detent spring seat	Gage
105. Height, detent spring seat to spherical surface	Gage
106. Radius, at perimeter of detonator cavity, max.	Gage
107. Position, detonator cavity with datum surface	Gage
108. Location, large flat	Gage
109. Damage	Visual 4/
110. Foreign matter	Visual
111. Angle, detent spring seat with large flat	Gage
112. Diameter, flash hole	Gage

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4.4.3.5 (Cont)

Categories and Defects	Method of Inspection
Minor:	
201. Width across mouth of detent spring seat	Gage
202. Web thickness	Gage
203. Position, diameter of detent spring seat with datum surface	Gage
204. Position, flash hole with datum surface	Gage
205. Radius, base of detent spring seat	Visual 3/
206. Detent spring seat encroaches upon flash hole	Visual
207. Sharp corner or edge	Visual/Feel
208. Finish	Visual/Feel

4.4.3.6 Booster holder (See Dwg. C7258872, a detail of Dwg. D7258864).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Pitch diameter, thread	Gage
102. Length, thread, min.	Gage 6/
103. Position, body diameter with datum surfaces	Gage
104. Diameter cavity	Gage
105. Position, cavity with datum surfaces	Gage
106. Length, underside of flange to top of crimping lip	Gage
107. Runout, spherical surface with datum surfaces	Gage
108. Thickness, web	Visual/Gage 5/
109. Length, flange to top of precision ball	Gage
110. Crack or split	Visual
111. Damage	Visual
112. Foreign matter	Visual
113. Diameter, crimping lip	Gage
Minor:	
201. Major diameter, thread	Gage
202. Diameter, flange	Gage
203. Thickness, flange	Gage
204. Thread undercut missing	Visual
205. Diameter, body	Gage
206. Location of thread	Gage 6/
207. Length, flange to closed end	Gage

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4.4.3.6 (Cont)

Categories and Defects	Method of Inspection
Minor (Cont)	
208. Radii, external, max. (except spherical radius)	Visual 3/
209. Finish	Visual/Feel
210. Protective coating damaged or missing	Visual
211. Burr, sliver, sharp corner or edge	Visual/Feel

4.4.3.7 Metal parts assembly (See Dwg. D7258864).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Length, shoulder to closed end, max.	Gage
102. Diameter, at nose cap crimp, max.	Gage
103. Exterior protective coating damaged or missing	Visual
104. Crack or split in body or nose cap	Visual
105. Foreign matter	Visual 2/
106. Nose cap crimp missing, improper, or less than 360°	Visual
107. Loose nose cap	Manual 7/
108. Damage	Visual 8/
109. Firing pin missing or not "free"	Visual/Manual 9/
110. Diameter of firing pin guide hole below firing pin body, min.	Gage
111. Depth to top of precision ball, min.	Gage
112. Contamination, cleaning agent residue, sealing compound, oil or grease	Visual
Minor:	
201. Length, shoulder to closed end, min.	Gage
202. Finish, exterior	Visual/Feel
203. Nose cap crimp below flush with body taper	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

2/ Particular attention should be given to cleanliness of the rotor detent spring groove. The presence of foreign matter of any nature in the groove shall cause the body to be classed defective. Machining chips are especially objectionable. The use of a bent probe or other suitable inspection aid is suggested to detect the foreign matter in the groove area.

3/ Components having doubtful compliance with the requirement(s) shall be gaged.

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4.4.3.7 (Cont)

4/ Raised metal protruding above the spherical surface shall be classed as damaged if the maximum spherical diameter, measured across the raised metal, is exceeded.

5/ A booster holder which complies with the specified dimensional requirements shall be classed defective if there is visual evidence of web rupture or perforation, as observed against an illuminated white background.

6/ These dimensions apply at the apex of the full formed threads.

7/ The assembly shall be classed defective if the nose cap turns on the body under thumb and finger torque.

8/ Thread deformity shall be classed as damage only if the assembly is rejectable by the "Go" thread gage.

9/ A sample unit having no firing pin, as determined by visual inspection, or having a firing pin which does not move freely within the firing pin guide hole, as determined by shaking and audible sound (or other approved method), shall be classed defective.

4.4.4 Tests.-The tests listed in Table I shall be performed on each lot in accordance with the applicable methods and procedures prescribed in 4.5. The samples shall be selected in such a manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

Table I

Test		Sample Size	Item	Req. Para.
Hardness	1/	50	Rotor detent spring	3.4
Corrosion resistance	2/	32	Booster holder	3.5.1.1
		32	Metal parts assembly	3.5.1.1

1/ If four or more units of the sample fail to comply with the hardness requirement, the lot shall be rejected. If two or three units fail to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples five or more units fail to comply with the hardness requirement, the lot shall be rejected.

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4.4.4 (Cont)

2/ The corrosion resistance test shall apply separately to the booster holder and metal parts assembly samples. If two or more units of the sample fail to comply with the corrosion resistance requirements, the lot shall be rejected. If one unit fails to comply with the requirements, a second sample consisting of the same number of units as specified for in the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirements, the lot shall be rejected.

4.4.5 Packaging, packing and marking inspection. -Inspection for packaging, packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.4.6 Inspection equipment. -Index of Inspection Equipment List (LI) 7258864 identifies the applicable Inspection Equipment Lists (IEL) required to perform examination and tests as specified herein. Equipment design(s) shall be in accordance with the applicable IEL code designations. The code designations are defined on Dwg. B11075228, a detail of LI 7258864. The provisions of MIL-A-2550 shall apply.

4.5 Test methods and procedures.

4.5.1 Hardness, rotor detent spring. -The method of test shall be as prescribed by the American Society for Testing and Materials (ASTM) E18, utilizing the appropriate Rockwell testing equipment and standard scale. The value of one impression taken at a point approximately midway in the widest area of the spring, shall apply as the hardness of the sample unit.

4.5.2 Corrosion resistance. -The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM B117, except that the finish shall not be disturbed before subjection to salt spray test.

5. PACKAGING.

5.1 Packaging and packing, Level C.-Each lot of fuze metal parts assemblies and components shall be packaged and packed in a container in such a manner as to be protected from corrosion and damage in accordance with MIL-STD-1169.

5.2 Marking and labeling.-Marking or labeling of packed shipping containers shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use.-These metal parts are intended for loading and assembly of the M505A3, P.D. fuze for various 20mm explosive projectiles when assembled with cartridge cases of the M103 or M21 type configuration.

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6.2 Ordering data.- Procurement documents should specify the following:

6.2.1 Title, number, and date of this specification.

6.2.2 Place of inspection, if not at the place of manufacture.

6.2.3 First article sample requirements (see 3.3, 4.3 and 6.3).

6.2.4 Packing and marking instructions (see 5.1 and 5.2).

6.2.5 Provisions for shipment and testing of first article samples.

6.2.6 Provision for the submission and approval of the manufacturing process (see 3.2).

6.3 First article sample.- Where a first article sample is required by the contract, the contracting officer will notify the designated inspection facility.

6.4 Process deviation.- A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.5 AQL's.- The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.6 Intermediate Point inspection.- The classification of defects identifies the defect characteristics (among other things) for acceptance inspection, It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

Custodians:

Army -MU
Air Force - 99

Preparing Activity:

Army - MU

Review Activities:

Army - MU
Air Force - 99

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE
MIL-F-46579D, Fuze, Point Detonating, M505A3; Parts For

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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